STUDENT TEAM WEEKLY REPORT

PROJECT / FEATURE	TEAM & MENTOR	WEEK / LEADER
Question Bank & Quiz Engine [Integration — Database, Backend & Frontend Integration]	TEAM ID : **2** FACULTY MENTOR:Mr. Rudragouda Patil	WEEK ENDING: 09/08/2026 STUDENT LEADER: Shruti Chavan.

WORK COMPLETED	CHALLENGES / RISKS	HELP NEEDED
• Database Schema Review: Reviewed and verified the ERD against Database.sql to ensure a consistent database structure. • PostgreSQL JDBC Integration: Configured and verified PostgreSQL database connectivity using JDBC. • REST API Documentation: Documented REST APIs for the Question Bank, Question, Quiz, and Quiz Attempt modules to support frontend integration. • Environment Setup: Configured the local PostgreSQL environment and JDBC database connection. • Integration Documentation: Created a consolidated Integration Layer README covering database setup, REST API references, integration workflow, and frontend integration guidelines.	• Frontend Integration: Frontend is not yet connected to the backend APIs, delaying end-to-end integration testing. • REST API Verification: Final verification of API request and response formats is pending until all modules are integrated. • Schema Consistency: Verified the ERD against Database.sql to ensure a consistent database structure. • End-to-End Testing: Complete integration testing will be performed after all project modules are fully connected.	• Need final verification of REST API endpoints before conducting end-to-end integration testing. • Need confirmation from the Frontend and Backend teams that API contracts and request/response formats are finalized.

INTEGRATION AND COLLABORATION	COMMON KPI EVIDENCES
Interfaces Used: Reviewed the Database.sql schema with the Database team and the Backend team's REST API documentation to verify integration requirements and API list**.** **Teams Consulted:** • Database team — Verified database schema, table relationships, and integration requirements. • Backend team — Reviewed REST API lists, service layer, and request/response formats. • Frontend team — Coordinated API consumption and data exchange requirements for frontend integration. **Pipeline Learnings:** Documented the complete Frontend → REST API → Business Logic → DAO → PostgreSQL workflow in the Integration Layer README for future development and onboarding.	• **Foundational Behavior:** Documented the Integration Layer architecture, REST APIs, and system workflow. • **Continuous Delivery:** Prepared and shared the Integration Layer README with setup instructions and API documentation. • **Automation:** Verified REST API lists and standardized JSON request and response formats for smooth integration.

NEXT WEEK COMMITMENTS	MENTOR RESPONSE
• Kartik Manoj: Complete frontend-backend API integration. • Anagha Thavkar: Perform end-to-end integration testing. • Shruti Chavan: Resolve API and database integration issues. • Pratiksha Chavan: Update integration documentation and API references.	

STUDENT TEAM WEEKLY REPORT

PROJECT / FEATURE	TEAM & MENTOR	WEEK / LEADER
Question Bank & Quiz Engine [Backend Logic Layer, Quiz Generator & Auto-Grading Engine]	**TEAM ID:** 2 **FACULTY MENTOR:** Mr. Rudragouda Patil	**WEEK ENDING:** 09/08/2026 **STUDENT LEADER:** Sumegh Bhamre

WORK COMPLETED	CHALLENGES / RISKS	HELP NEEDED
• **Core Logic Services:** Implemented QuestionBankService, QuestionService, and QuestionOptionService managing bank lifecycles (DRAFT → SUBMITTED → APPROVED / REJECTED → LIVE). • **Fisher-Yates Quiz Randomizer:** Developed O(N) unbiased random exam paper creation in QuizServiceImpl filtering questions by Subject, Topic, and BTL level. • **Instant Auto-Grading Engine:** Built QuizAttemptServiceImpl using O(1) Hash Table lookups for automatic MCQ grading and score tracking. • **Moderator Review Workflow:** Enabled inline stem, BTL, mark, and option key editing during review before approval. • **Evidence & Repo:** Codebase sanitized and pushed to GitHub branch *logic_layer_sumegh*.	• **Invalid MCQ Options:** Risk of zero/multiple correct choices. *Action:* Built QuestionOptionValidator enforcing 4 choices & 1 correct choice. **GREEN (RESOLVED)** • **Moderation Lock Conflicts:** Typos caused total bank rejections. *Action:* Allowed edits in SUBMITTED state while locking APPROVED banks. **GREEN (RESOLVED)** • **Duplicate Exam Attempts:** Risk of re-taking exams. *Action:* Added attempt locking (COMPLETED state) and SQL ON CONFLICT DO UPDATE. **GREEN (RESOLVED)**	**Technical Guidance Request:** Could the integration team check whether we have written logic's for everything that is going to be there on frontend?

INTEGRATION AND COLLABORATION

- **Interfaces Used:** Connected Java DAOs to PostgreSQL schema (Database.sql).
- **Teams Consulted:**
  - Collaborated with Database team for userId, user role (TEACHER, MODERATOR, STUDENT), and 2-column subject binding.
  - Database team consulted to review the schema and discuss if there are any changes.
  - Integration team consulted to finalize the branch creation, so that they can work efficiently.
  - Front end team consulted for what buttons they would be adding.
- **Pipeline Learnings:** Gained end-to-end data pipeline insights linking Database → DAO Service.

COMMON KPI EVIDENCES

- **Foundational Behavior:** Quiz generation execution latency < 15ms (O(N) Fisher-Yates execution).
- **Continuous Delivery:** Clean Git commit history on branch *logic_layer_sumegh*. Sanitized codebase free of secrets or binaries.
- **Automation:** 100% Automation in Question bank creation and Moderator review.

NEXT WEEK COMMITMENTS	MENTOR RESPONSE
• **Sumegh Bhamre:** Support Frontend Team with the exploration of more designs to enhance user experience. *(Due: Aug 12, 2026)* • **Sumegh Bhamre:** Execute end-to-end integration testing with Frontend UI & Auth modules. *(Due: Aug 14, 2026)*	

STUDENT TEAM WEEKLY REPORT

PROJECT / FEATURE	TEAM & MENTOR	WEEK / LEADER
Question Bank & Quiz Engine [Frontend UI Architecture, Role Dashboards & Auth Flow]	**TEAM ID:** 2 **FACULTY MENTOR:** Mr. Rudragouda Patil	**WEEK ENDING:** 09/08/2026 **STUDENT LEADER:** Srushti Salgar

WORK COMPLETED	CHALLENGES / RISKS	HELP NEEDED
Multi Role Authentication & Navigation: Built SignIn and Self-Registration UI supporting 4 roles (`Admin`, `Author`, `Moderator`, `Student`), complete with client-side form validation and quick-fill demo profiles. **Question Bank Authoring Pipeline:** Designed a 2-step setup for MCQ and Descriptive questions, supporting manual entry, CSV uploads with template downloads, and staged batch submissions . **Faculty Moderation Workspace :** Implemented an inline review queue for moderators to inspect question banks, edit options, add feedback notes, and update approval states. **Admin Dashboard UI:** Designed, styled, and integrated the `admin` view layout with responsive navigation, system overview panels, and role-management interfaces.	**Authentication Module:**Successfully developed responsive Sign In and Registration pages with client-side form validation and role selection. **Status: GREEN** **Responsive UI Design:**Designed a consistent and responsive user interface that works across different screen sizes and devices. **Status: GREEN** **Role-Based Frontend Structure:** Established separate frontend modules for Administrator, Student (Quiz Taker), Question Author, Moderator, and Instructor to support role-specific functionality. **Status: GREEN**	**Database Setup:** Require database team support for PostgreSQL integration and queries. **Backend APIs:** Need REST endpoints to connect frontend views to the server.

INTEGRATION AND COLLABORATION

- **Interfaces Used:** HTML5, CSS Variables, JavaScript, Google Fonts API.
- **Teams Consulted:**
  - Consulted logic layer team for aligned UI status transitions with backend logic for consistent author, moderator, and student views.
  - . Standardized UI data schemas for eventual integration for feature branch.

COMMON KPI EVIDENCES

- UI loads in under 12ms with a fast, stable layout across desktop screens.
- Completed, styled, and pushed all primary user interface layouts to the team repository.
- Automated file parsing, timed exam sessions, and instant result generation with zero script crashes

NEXT WEEK COMMITMENTS	MENTOR RESPONSE
Execute joint end to end integration testing and the Integration Team to validate system data flow. Replace `localStorage` mock handlers with live `fetch()` API calls for Auth (`/auth/login`) and Question Authoring	

STUDENT TEAM WEEKLY REPORT

PROJECT / FEATURE	TEAM & MENTOR	WEEK / LEADER
Question Bank & Quiz Engine [Database Design & Initial Data Preparation]	**TEAM ID:** 2 **FACULTY MENTOR:** Mr. Rudragouda Patil	**WEEK ENDING:** 09/08/2026 **STUDENT LEADER:** Varad Tanawade

WORK COMPLETED	CHALLENGES / RISKS	HELP NEEDED
• Designed normalized database schema. • Established PK/FK relationships. • Finalized schema. • Collected Student, Faculty, Descriptive and MCQ CSV datasets. • Mapped CSV fields to database. • Planned Java CSV import utility. Evidence: SQL schema, ER mapping and datasets.	• Amber: Different CSV formats require preprocessing. • Amber: Possible duplicates/missing values. • Green: Schema finalized. • Green: Initial mapping completed.	**Technical Guidance Request:** Need review on the preferred Java/JDBC approach for CSV-to-database import.

INTEGRATION AND COLLABORATION

- Finalized schema with team.
- Coordinated dataset collection.
- Created Structured ER Diagram for schema visualization.
- Discussed end-to-end import workflow.
- Planned backend integration.

COMMON KPI EVIDENCES

- Foundational Behavior: Normalized database schema , ensuring minimal redundancy and referential integrity across all entities.
- Continuous Delivery: Database schema versioned in Git with documented SQL scripts and coordinated with the backend team for seamless API integration. Creation of step by step brief readme for database initialization
- Automation: Database structure prepared for 100% automated population through the existing CSV extraction module, supporting Student, Faculty, Descriptive Question Bank, and MCQ Question Bank datasets.
- Collaboration: Successfully aligned database schema with backend APIs and validated entity mappings to support end-to-end quiz generation workflow.

NEXT WEEK COMMITMENTS	MENTOR RESPONSE
• Testing Java import for Student & Faculty CSVs. • Testing Java import for Descriptive & MCQ CSVs.. • Team – Integrate backend. • Covert database to higher NF forms	